Pilot-Scale Text Column Detection and Reading-Order Recovery for Traditional Mongolian Historical Archives

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Abstract

Traditional Mongolian historical archives pose a challenging page-level document analysis problem because the script is vertically written, cursive, low-resource, and frequently degraded by aging paper, ink variation, seals, scanning rotation, and heterogeneous archive sources. Since expert transcription of handwritten traditional Mongolian columns is expensive and currently unavailable, this concise study focuses on an upstream stage for archival OCR: text-column detection and reading-order recovery. We construct a page-level annotation protocol with text-column bounding boxes, ignored noise regions, orientation metadata, and reading order. We compare classical rule-based extraction, YOLOv8n, and a COCO-pretrained Faster R-CNN MobileNet detector with a rotation-aware ordering strategy. On an expanded 54-page test set, YOLOv8n improves the project-level IoU@0.5 F1 score from 0.722 for the heuristic baseline to 0.875 and achieves 0.887 reading-order accuracy on matched columns, while Faster R-CNN reaches 0.866 F1. However, full-page success remains low (0.074 for YOLOv8n), so the system should be interpreted as a candidate crop generator rather than a fully automatic solution. We also conduct fair-budget retraining of transfer baselines (RT-DETR-L, DocLayout-YOLO) and find that DocLayout-YOLO improves from F1 = 0.037 (5 epochs) to 0.602 (50 epochs), demonstrating that modern layout priors can transfer with sufficient domain adaptation. We provide a crop-export-ready interface, but do not report CER/WER because expert column-level transcripts are not yet available.

Keywords: Historical document analysis, Traditional Mongolian archives, Text column detection, Reading order recovery, Low-resource document analysis

1. Introduction

Full-page recognition of traditional Mongolian historical archives requires more than a recognizer trained on isolated text lines. Pages often contain many vertical handwritten columns, irregular spacing, red seals, page borders, paper stains, and inconsistent scan orientations. If columns are not localized and ordered reliably, downstream OCR modules receive incomplete or misordered crops.

In the current project stage, expert transcription of handwritten traditional Mongolian archive columns is not available. Rather than reporting unverifiable OCR accuracy, this paper evaluates the upstream page-layout problem using bounding-box and reading-order annotations. Layout annotation is cheaper than full transcription and provides the bridge between scanned archive pages and later recognition modules.

Our contributions are threefold. First, we define a page-level annotation protocol for traditional Mongolian archives, including text-column boxes, ignore regions, orientation, and reading order, with refined Ignore subtypes for real-world archival artifacts. Second, we provide a pilot-scale evaluation of rule-based and learning-based column detection under a unified matching protocol, with bootstrap confidence intervals and fair-budget transfer baseline comparison. Third, we implement a crop-export-ready interface while clearly separating layout evaluation from future transcription-based OCR evaluation. The work is positioned as an application-oriented diagnostic study rather than a new detector architecture.

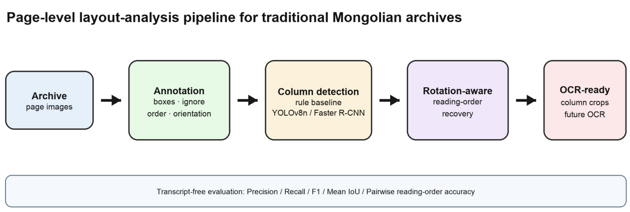

Figure 1: Overview of the page-level layout-analysis pipeline.

2. Related Work

Historical document layout analysis. Segmentation remains a prerequisite for reliable historical document recognition. Surveys and historical datasets show that degradation,

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background noise, and handwriting variability make page decomposition difficult [1, 2, 3]. Large-scale multi-style historical benchmarks such as M5HisDoc [4] have advanced evaluation protocols for historical inputs. General-purpose historical document segmentation frameworks such as dhSegment [5] and datasets such as DIVA-HisDB [6] have enabled reproducible benchmarks; ICDAR RDCL [7] systematizes complex layout evaluation. Traditional Mongolian recognition has been studied at word or character level, including MOLHW [8] and CNN-Transformer hybrids for Mongolian ancient documents [9]. A Mongolian movable-type newspaper dataset [10] highlights the persistent data scarcity. Nevertheless, page-level layout analysis for *handwritten* historical Mongolian archives remains underexplored.

Learning-based document detection. PubLayNet, DocLayNet, and LayoutParser have promoted reusable learning-based document-layout pipelines [11, 12, 13]. Recent YOLO-style document-layout models also show competitive detection ability in modern document settings [16, 17]. These resources mainly target modern printed or PDF-derived pages, whereas our archive pages are handwritten, vertically arranged, and affected by scan rotation and cultural-heritage degradation.

Reading-order recovery. Reading-order recovery is necessary for converting detected regions into structured text. Handwritten document understanding still requires ordering beyond line recognition [14], and historical OCR frameworks treat layout analysis, segmentation, and recognition as connected stages [15]. We evaluate column detection and order recovery independently, and export crops for future recognition once expert transcripts become available.

3. Dataset and Annotation Protocol

The dataset contains 105 valid traditional Mongolian archive pages after removing Chinese handwritten archive pages and correcting abnormal scan orientation. The split contains 43 training pages, 8 validation pages, an original 11-page test split, and a 43-page test-expansion split, giving an expanded 54-page test set. The annotations include 1538 valid text-column boxes and 199 ignored regions.

Table 1: Page-level annotation statistics.

Split	Pages	Columns	Ignored
train	43	637	67
val	8	115	10
original test	11	152	45
expanded test	43	634	77
total	105	1538	199

Each non-ignored text column is annotated with a bounding box and reading-order index. Ignored regions cover six subtypes: seals or stamps, stains and background noise, page borders or scanning edges, non-target-language regions, ambiguous fragments, and unseparable artifact–text overlaps. A 5-page independent IAA subset gives text-column box F1@0.5 of 0.852 and mean matched IoU of 0.800; reading-order agreement is lower at 0.677, and Ignore F1@0.5 of 0.286 is acknowledged as a limitation.

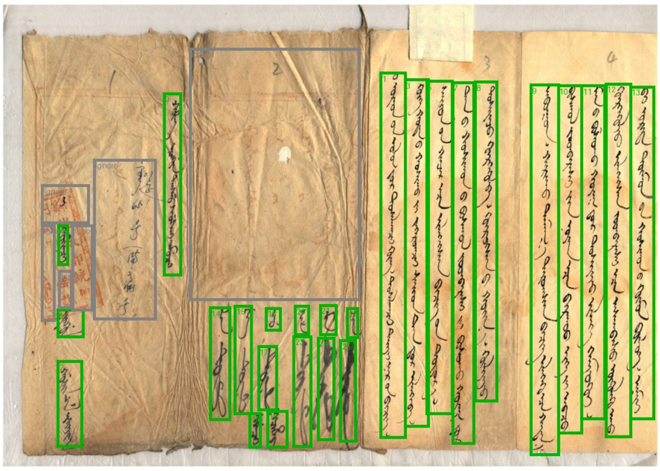

Figure 2: Example of the annotation protocol. Green boxes indicate valid text columns; gray boxes indicate ignored regions.

4. Method

Rule-based baseline. The heuristic baseline applies adaptive binarization, vertical ink-density projection, connected-component filtering, and width, height, and density constraints, followed by overlap merging and left-to-right order assignment.

Learning-based detectors. We train YOLOv8n [18] for 50 epochs with image size 960 and batch size 1, initialized from public weights. The confidence threshold 0.35 is selected only on the 8-page validation split. We also fine-tune a COCO-pretrained Faster R-CNN MobileNetV3-FPN detector for 5 epochs with max-side 960; threshold 0.80 is selected on validation.

Transfer baselines with fair-budget retraining. RT-DETR-L and DocLayout-YOLO are included as diagnostic transfer experiments, each at two training budgets: 5 epochs (matching the Faster R-CNN budget) and 50 epochs (matching YOLOv8n). The 50-epoch retraining tests whether modern document-layout priors transfer to this domain under fair optimization effort.

Rotation-aware reading order. For detections $b_i = (x_{1i}, y_{1i}, x_{2i}, y_{2i})$, the column center is $c_i = (x_{1i} + x_{2i})/2$. Detections are sorted by c_i ascending for ordinary pages and descending for rotated pages. We note that the pairwise reading-order metric is computed only on matched detection–ground-truth pairs, making it a proxy for the ordering module rather than a fully independent metric.

Crop-export interface. The pipeline exports oracle crops and detected crops for future OCR evaluation. No CER/WER is reported.

5. Experiments

5.1. Metrics

Column detection is evaluated with greedy IoU@0.5 matching. We report precision, recall, F1, mean matched IoU, and pairwise reading-order accuracy on matched columns. Predictions substantially overlapping ignored regions are not counted as false positives. We also report complete detection rate, full-page success rate, and page-level bootstrap 95% confidence intervals (1000 samples).

5.2. Main Results (Expanded 54-page test)

Table 2: Expanded 54-page test-set comparison under IoU@0.5 matching.

Method	Prec.	Rec.	F1	mIoU	RO
Heuristic	0.694	0.753	0.722	**0.881**	0.882
YOLOv8n	**0.936**	**0.821**	**0.875**	0.756	**0.887**
FRCNN	0.924	0.816	0.866	0.744	0.870

Table 2 reports results on the expanded 54-page test set (786 GT columns). Both learned detectors achieve substantially higher F1 than the heuristic baseline. YOLOv8n and Faster R-CNN are close in F1, with overlapping bootstrap intervals (Table 3).

Table 3: Strict page-level metrics and bootstrap 95% CI on the expanded 54-page test.

Method	Complete	Full-page	F1 95% CI
Heuristic	0.148	0.148	[0.643,0.800]
YOLOv8n	0.111	0.074	[0.834,0.906]
FRCNN	0.111	0.093	[0.830,0.897]

Full-page success remains low for all methods, because even one missed or extra column makes complete detection fail.

5.3. Transfer Baseline Results

Table 4: Transfer baselines at two training budgets on the original 11-page test split.

Method	Prec.	Rec.	F1	mIoU	RO
RT-DETR 5ep	0.032	0.270	0.058	0.632	0.847
RT-DETR 50ep	0.183	0.132	0.153	0.622	1.000
DocLayout 5ep	0.046	0.031	0.037	0.572	0.000
DocLayout 50ep	0.635	0.572	0.602	0.709	0.900

Table 4 reports transfer baselines at two budgets on the 11-page diagnostic split. DocLayout-YOLO improves from F1 = 0.037 to 0.602 (16×), showing that DocStructBench priors contain transferable features with sufficient domain adaptation. RT-DETR-L improves but remains far below competitive levels (F1 = 0.153). Due to computational limits, fair-budget baselines have not been re-evaluated on the expanded test.

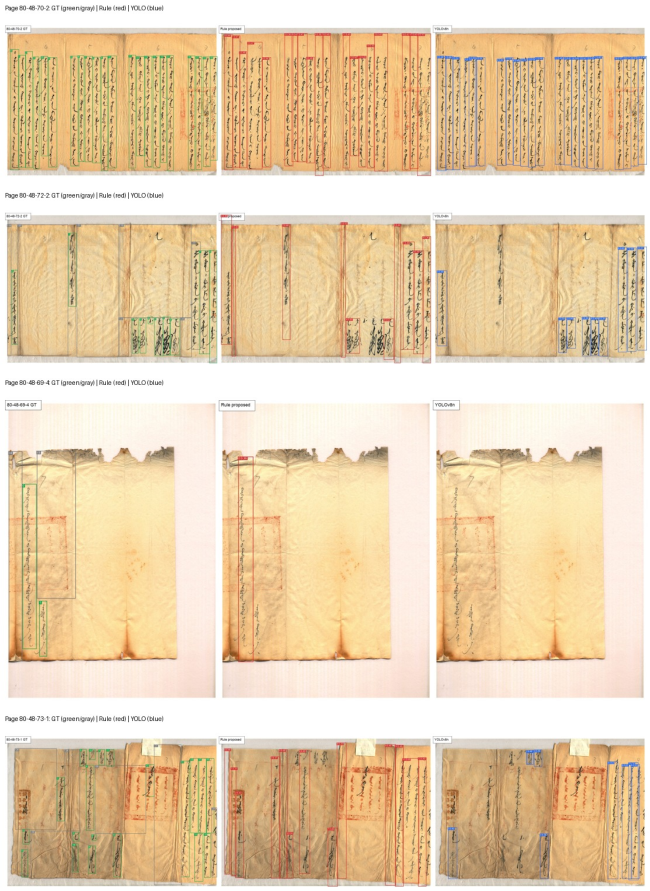

Figure 3: Representative comparisons. Green: valid text columns, gray: ignored, red: rule-based, blue: YOLO.

5.4. Stratified Analysis

Table 5 reports YOLOv8n stratified performance on the expanded 54-page test. Many-column pages achieve the best F1 (0.907), while few-column pages (F1 = 0.720) and pages with Ignore regions (F1 = 0.808) remain more challenging. The expanded stratification is more informative than the original 11-page diagnostic because subgroup sizes are larger.

Table 5: Stratified YOLOv8n on the expanded 54-page test.

Group	Pages	Prec.	Rec.	F1
Few (≤5)	4	0.900	0.600	0.720
Medium (6–15)	22	0.869	0.727	0.791
Many (>15)	28	0.936	0.880	**0.907**
With Ignore	7	0.924	0.718	0.808
No Ignore	47	0.914	0.837	0.874

Additional diagnostics (IoU@0.75, training-set scaling, threshold sensitivity, column-aware post-processing, ignore-region and rotation-order ablation, runtime, YOLOv8n multi-scale retraining at 640/960/1280) are in the supplementary material. Multi-scale retraining confirms image size 960 yields the best test F1 (0.888).

6. Discussion and Limitations

The main value of this concise study is a realistic pilot-scale evaluation protocol for a low-resource archival layout problem, not a complete OCR system. The expanded 54-page test set reduces statistical uncertainty compared with the original 11-page test, but the dataset remains modest and bootstrap intervals should be interpreted as honest uncertainty estimates. All hyper-parameters are selected on a validation split of only 8 pages, introducing potential overfitting to the validation distribution.

The transfer baseline experiments demonstrate that fair-budget comparison is essential: DocLayout-YOLO's F1 improved from 0.037 to 0.602 simply by matching YOLOv8n's training budget. Faster R-CNN was trained for only 5 epochs, and a compute-matched comparison with YOLOv8n is left to future work. Ignore-region agreement remains weak (F1@0.5 = 0.286). Full-page success remains low, so future work should measure human correction time and downstream OCR effects.

7. Conclusion

We presented a pilot-scale page-level layout-analysis evaluation protocol for traditional Mongolian historical archives. On an expanded 54-page test set, YOLOv8n achieves F1 = 0.875 vs. 0.722 for the heuristic baseline, while Faster R-CNN reaches 0.866. Fair-budget retraining reveals that DocLayout-YOLO can improve from F1 = 0.037 to 0.602 with sufficient domain adaptation. However, full-page success remains low (0.074) and no new detector architecture or learned reading-order module is proposed. The present system should therefore be viewed as a candidate crop generator and diagnostic evaluation package for future traditional Mongolian archival OCR research.

Data Availability

Non-image research artifacts (annotation schema, 105-page split manifest, prediction JSON files, Python evaluation scripts, training protocol documentation including random seed 20260513, optimizer SGD momentum 0.937, and environment versions Python 3.9, PyTorch 2.8, ultralytics 8.4) are prepared for release. Archive images are subject to institutional access constraints; metric recomputation from shared annotations and prediction files is fully reproducible without retraining or GPU access.

Acknowledgments

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